

Transforming [^{177}Lu]Lu-PSMA-617 treatment planning: Machine learning-based radiodosimetrics and swin UNETR using pretherapy PSMA positron emission tomography/computed tomography (PET/CT)

Elmira Yazdani¹ | Aryan Neizehbaz² | Najme Karamzade-Ziarati³ |
Farshad Emami⁴ | Habibeh Vosoughi³ | Mahboobeh Asadi³ | Atefeh Mahmoudi¹ |
Mahdi Sadeghi¹ | Saeed Reza Kheradpisheh² | Parham Geraimifar³

¹Department of Medical Physics, School of Medicine, Iran University of Medical Sciences, Tehran, Iran

²Department of Computer and Data Sciences, Faculty of Mathematical Sciences, Shahid Beheshti University, Tehran, Iran

³Research Center for Nuclear Medicine, Tehran University of Medical Sciences, Tehran, Iran

⁴Nuclear Medicine and Molecular Imaging Department, Imam Reza International University, Razavi Hospital, Mashhad, Iran

Correspondence

Mahdi Sadeghi, Iran University of Medical Sciences (IUMS), Hemat Highway, next to Milad Tower, Tehran, Iran.
Email: sadeghi.m@iums.ac.ir

Saeed Reza Kheradpisheh, Shahid Beheshti University (SBU), Shahid Shahrari Square, Evin, Tehran, Iran.
Email: sr.kheradpisheh@gmail.com

Parham Geraimifar, Tehran University of Medical Sciences (TUMS), Kargar St, District 6, Pour Sina St, Tehran, Iran.
Email: pgeramifar@tums.ac.ir

Funding information

School of Medicine, Iran University of Medical Sciences, Grant/Award Number: 25160

Abstract

Background: Personalized pretreatment dosimetry planning is crucial for optimizing [^{177}Lu]Lu–prostate-specific membrane antigen-617 (Lu-PSMA) radioligand therapy (RLT) in patients with metastatic castration-resistant prostate cancer (mCRPC).

Purpose: This study addresses two goals. First, we develop a machine learning (ML)-based pretreatment planning model to predict post-therapy absorbed doses (ADs) in metastatic lesions by integrating clinical biomarkers (CBs) with radiomic features (RFs) and dosiomic features (DFs) extracted from [^{68}Ga]Ga-PSMA-11 (Ga-PSMA) positron emission tomography/computed tomography (PET/CT), thereby improving predictive accuracy. Second, we develop a transformer-based deep learning (DL) architecture to predict Monte Carlo (MC)-derived dose rate maps (DRMs), minimizing reliance on computationally intensive MC simulations.

Methods: For the ML objective, retrospective posttreatment dosimetry data from 20 patients with mCRPC treated with Lu-PSMA RLT were used as ground truth labels. Patient-specific MC dosimetry was employed on Ga-PSMA PET/CT

Abbreviations: AD, Absorbed dose; AdaBoost, Adaptive Boosting; ALP, Alkaline phosphatase; B-H, Benjamini-Hochberg; BM, Bone metastases; Bq, Becquerel; CA, Calcium; CBS, Clinical biomarkers; CNN, Convolutional neural network; CR, Creatinine; CT, Computed tomography; DD, Dose difference; DF, Dosiomic feature; DL, Deep learning; DPK, Dose point kernel; DRM, Dose rate map; DTA, Distance to agreement; EBRT, External beam radiotherapy; Ga-PSMA, [^{68}Ga]Ga-PSMA-11; GATE, Geant4 application for emission tomography; GBR, Gradient boosting regressor; GLCM, Gray level co-occurrence matrix; GLRLM, Gray level run-length matrix; GLSZM, Gray level size zone matrix; GS, Gleason score; HCT, Hematocrit; HU, Hounsfield units; IBSI, Image biomarker standardization initiative; LASSO, Least absolute shrinkage and selection operator; LIME, Local interpretable model-agnostic explanations; LNM, Lymph node metastases; LOOCV, Leave-one-out cross-validation; Lu-PSMA, Lutetium-177-labeled prostate-specific membrane antigen; MAE, Mean absolute error; MC, Monte Carlo; mCRPC, Metastatic castration-resistant prostate cancer; MI, Mutual information; ML, Machine learning; MSE, Mean squared error; NET, Neuroendocrine tumor; NGTDM, Neighboring gray tone difference matrix; OSEM, Ordered-subset expectation maximization; P, Phosphor; PET, Positron emission tomography; PSA, Prostate-specific antigen; PVE, Partial volume effect; RF, Radiomic feature; RFE, Recursive feature elimination; RFR, Random forest regressor; RMSE, Root mean squared error; SD, Standard deviation; SHAP, Shapley additive explanations; SPECT, Single-photon emission computed tomography; SSL, Self-supervised learning; ST, Soft tissue; SW-MSA, Shifted-window multi-head self-attention; Swin UNETR, Shifted windows UNET transformers; TIAC, Time-integrated activity coefficient; TRT, Targeted radionuclide therapy; VOI, Volume of interest; VSV, Voxel S-value; W-MSA, Window-based multi-head self-attention; XAI, Explainable artificial intelligence; XGBR, EXtreme gradient boosting regressor.

images using the GATE v9.1 toolkit to generate DRMs. After image preprocessing, RFs and DFs were extracted from Ga-PSMA CT images and DRMs using LIFEx v7.4.0. Multiple feature selection techniques, including recursive feature elimination (RFE), mutual information, Boruta, LASSO, and Elastic Net, were applied and evaluated. The Benjamini-Hochberg correction ($q < 0.05$) was used to control for false discovery rate following each method. Multiple nonlinear regression models were trained using leave-one-out cross-validation (LOOCV), and model interpretability was assessed using SHAP and LIME radar plots. A shifted windows UNET Transformers (Swin UNETR) architecture with self-supervised learning (SSL) pretraining was employed to predict voxel-wise PET-based DRMs for the DL objective. The model was fine-tuned on MC-labelled DRM data from 30 patients (including 10 additional cases) using 5-fold cross-validation.

Results: Among multiple feature selection strategies, RFE was ultimately selected for final modelling based on its superior predictive performance. The ensemble tree regressor (ETR) using selected CT RFs, PET DFs, and significant CBs achieved an $R^2 = 0.82$ and RMSE = 0.67 Gy/GBq. For DRM prediction, the SSL-pretrained Swin UNETR achieved an R^2 of 0.97, NRMSE of 0.003 Gy/GBq, and a Gamma pass rate of 99.08%, closely matching MC-derived DRMs.

Conclusions: Integrating ML-based radiodosimetrics and transformer-based DL enables accurate, efficient lesion AD and DRM prediction from pretherapy PET/CT, supporting personalized Lu-PSMA RLT planning.

KEYWORDS

dose prediction, machine/deep learning, mCRPC, radiomics and dosimetrics, radiopharmaceutical therapy, self-supervised learning, swin UNETR, treatment planning

1 | INTRODUCTION

Lutetium-177-labeled prostate-specific membrane antigen (Lu-PSMA) targeted radionuclide therapy (TRT), also known as radiopharmaceutical therapy (RPT), has emerged as a promising therapeutic approach in radiation oncology for patients with metastatic castration-resistant prostate cancer (mCRPC).¹ However, optimizing the balance between therapeutic absorbed dose (AD) and treatment-associated toxicity remains a significant challenge.^{2,3} Traditional dosimetry methods often overlook interpatient variability, highlighting the need for advanced predictive modelling to enhance individualized treatment planning and clinical outcomes.⁴

Monte Carlo (MC) simulations, considered the gold standard, provide unparalleled precision in dose calculation.^{2,3,5} However, their substantial computational demands limit widespread clinical use. To improve accuracy and efficiency, alternative strategies have been proposed, including machine learning (ML) models and deterministic techniques such as the dose point kernel (DPK) and voxel S-value (VSV) methods.⁶ Evidence suggests that pretherapeutic scans can predict posttreatment ADs or correlate pre- and posttreatment imaging in patients with mCRPC.^{6–8} Additionally, deep learning (DL) has emerged as a promising tool for tailoring dose distribution in TRT using pretherapy positron

emission tomography/computed tomography (PET/CT) imaging with Ga-68 or F-18 tracers.^{6,7,9}

Radiomics is a paradigm in imaging and precision medicine that aids cancer prognosis and lesion characterization.^{6,10,11} Dosimetrics, which analyses spatial dose patterns, is well-studied in external beam radiotherapy (EBRT) but remains less explored in RPT.^{12,13} Recently, integrating radiomics and dosimetrics (radio-dosimetrics) has gained interest for improving treatment response assessment and prognosis.¹⁴ This approach has also been applied to TRT, where an ML-based radio-dosimetrics model using [⁶⁸Ga]Ga-DOTATOC PET/CT images and dose rate maps (DRMs) was developed to predict ADs in organs for neuroendocrine tumours (NETs) treated with [¹⁷⁷Lu]Lu-DOTATATE. The study suggested that selected features could enhance treatment safety and efficacy.¹⁵

We previously developed an ML model that integrated clinical biomarkers (CBs) with radiomic features (RFs) extracted from [⁶⁸Ga]Ga-PSMA-11 (Ga-PSMA) PET/CT images to predict ADs in the kidneys and metastatic lesions of 20 patients with mCRPC undergoing Lu-PSMA TRT. The model performed well for kidneys (RMSE: 0.11 Gy/GBq, R^2 : 0.87), but for lesions, the best results (using selected PET/CT RFs and all significant CBs) achieved an R^2 of 0.77 and an RMSE of 1.04 Gy/GBq, indicating the need for further refinement. Assuming that DRM-based PET provides more

meaningful features than standard PET images, the current study incorporates dosiomic features (DFs) from Ga-PSMA DRMs to enhance predictive accuracy for metastatic lesions. The ground truth Ga-PSMA DRMs were derived from patient-specific MC-based dosimetry using the Geant4 application for emission tomography (GATE) simulation.

Building on this approach, this study has two main objectives. First, it aims to develop an explainable ML-based model for predicting ADs in metastatic lesions of patients with mCRPC undergoing Lu-PSMA TRT. The model integrates CBs with RFs from Ga-PSMA CT images and DFs from Ga-PSMA DRMs, which may provide more informative data than PET images alone. However, generating MC-based PET DRMs is time-consuming, and obtaining a sufficiently large dataset is challenging. Dose assessment is generally based on sequential single-photon emission computed tomography (SPECT) imaging at various time points over several days following each treatment cycle. In routine clinical practice, obtaining extensive imaging data is rarely feasible, and post-therapy dosimetry is performed on only a few patients, resulting in insufficient datasets for training DL models.¹⁶ These limitations pose significant challenges in developing and validating robust DL solutions for treatment planning and dosimetry.⁹

To address these challenges, this study employed self-supervised learning (SSL) with shifted windows UNET Transformers (Swin UNETR) for personalized dosimetry using Ga-PSMA pretherapy images. Transformers enable efficient parallel processing and perform well with limited data. Previously, we developed an SSL model for Ga-PSMA PET/CT segmentation, training on 652 unlabelled images and fine-tuning on 100 labelled ones.¹⁷ In the present study, we leveraged its pretrained weights and refined them on labelled DRM images to improve dosimetry prediction. Following this, the AD distributions generated by DL were validated by comparing them with those obtained from MC-based simulations. This approach enhances treatment assessment and informs dose escalation strategies.

To promote transparency and reproducibility, the complete source code for both the ML and DL pipelines, including preprocessing, feature extraction, and model implementation, is publicly available on GitHub (<https://github.com/ElmiraYazdani/Machine-Learning-Based-Radiodosimics-and-Swin-UNETR-Framework-for-Dose-Prediction>).

2 | MATERIALS AND METHODS

2.1 | Patient cohort

For the first goal, this study utilized a cohort similar to that in our previous research, including 20 patients with mCRPC (68.28 ± 6.44 years) with PSMA-expressing

lesions visible on Ga-PSMA PET/CT within 2 months of Lu-PSMA RLT.¹⁸ All patients provided informed consent as part of an IRB-approved study. The net administered dose of RLT was 6.8 ± 0.4 GBq, calculated by subtracting residual activity from the original value. Scintigraphy and SPECT/CT imaging were acquired, and blood samples were collected prior to RLT for CBs. The CBs for the 20 patients with mCRPC in this study were similar to those reported in Table 1 of our previous publication, including demographic information (age, weight, and height), previous therapies, laboratory test results (e.g., prostate-specific antigen [PSA], creatinine [CR], alkaline phosphatase [ALP], hematocrit [HCT], calcium [CA], and histopathological data such as Gleason score [GS]).

For the second task, which involves generating DRM using DL, we expanded the cohort by adding 10 additional patients, bringing the total number of patients used for DL-based DRM generation to 30. This additional data helps improve the DL model's performance by increasing the diversity of the training dataset. Unlike the first task, this goal does not require Lu-PSMA treatment and focuses instead on generating PET-based DRMs from Ga-PSMA images.

2.2 | Imaging and dosimetry

Our previous work comprehensively describes the imaging protocols, calibration, image segmentation, and Lu-PSMA MC-based dosimetry procedure.¹⁸ Briefly, PET/CT acquisitions were performed using the Siemens Biograph 6 TruePoint scanner, employing ordered-subset expectation maximization (OSEM) reconstruction with a 168×168 matrix and a pixel size of $4.072 \text{ mm} \times 4.072 \text{ mm}$. A low-dose CT scan was also acquired for attenuation correction. For Lu-PSMA RLT dosimetry, serial post-therapy scintigraphy was conducted at 4, 48, and 72 h, with a quantitative SPECT/CT scan at 48 h. SPECT scans were acquired on the Siemens Symbia T2 system, utilizing OSEM reconstruction and attenuation correction with low-dose CT scans, resulting in a 128×128 matrix and an isotropic resolution of 4.79 mm^2 . Detailed imaging parameters are provided in Table 2 of our previous work, which outlines the SPECT/CT and scintigraphy imaging acquisition protocols. A calibration technique derived the SPECT camera factor [cps/MBq] from the ratio of total counts to scan duration and net activity, converting SPECT images from counts to Becquerel (Bq) units.¹⁸

Segmentation was done using 3D Slicer v5.2.2 by a certified nuclear medicine physician, focusing on the largest eight lesions per patient and excluding those $< 4 \text{ mL}$ to reduce partial volume effect (PVE). Manual segmentation on high-resolution PET/CT images was co-registered with SPECT/CT images for accuracy. Lesions were categorized as bone metastases

TABLE 1 Selected features from three experiments for lesion absorbed dose (AD) prediction based on the recursive feature elimination (RFE) feature selection method.

Experiment	Selected variable names and numbers ($q < 0.05$)
E1: selected CT RFs and DRM DFs	10: GLSZM_ZoneSizeNonUniformity_DRM, GLSZM_NormalisedZoneSizeNonUniformity_DRM, GLCM_InverseDifference_DRM, GLCM_SumEntropy_DRM, GLSZM_SmallZoneLowGreyLevelEmphasis_DRM, GLSZM_GreyLevelVariance_DRM, GLRLM_ShortRunsEmphasis_DRM, GLSZM_SmallZoneHighGreyLevelEmphasis_CT, GLSZM_ZonePercentage_CT, and GLSZM_ZoneSizeNonUniformity_CT
E2: E1 + selected CBs	12: E1, PSA, and ALP
E3: E1 + whole significant CBs ($q < 0.05$)	16: E1, PSA, ALP, CA, P, HCT, and CR

***Abbreviations:** ALP, alkaline phosphatase; CA, calcium; CB, clinical biomarker; CR, creatinine; DF, dosimetric feature; DRM, dose rate map; GLCM, gray Level Co-occurrence Matrix; GLRLM, gray level run length matrix; GLSZM, gray level size zone matrix; HCT, hematocrit; P, phosphor; PSA, prostate-specific antigen; RF, radiomic feature.

TABLE 2 Best predictive algorithms for each experiment, with corresponding metrics based on the recursive feature elimination (RFE) feature selection method.

Experiment	Best model	R ² Score	MAE (Gy/GBq)	MSE (Gy/GBq)	RMSE (Gy/GBq)	Spearman's rank Correlation ($q < 0.05$)
E1	XGBR	0.74	0.78	1.47	0.77	0.70
E2	XGBR	0.79	0.72	1.2	0.72	0.76
E3	ETR	0.82	0.68	1.03	0.67	0.79

***Abbreviations:** ETR, extra tree regressor; MAE, mean absolute error; MSE, mean squared error; R² Score, coefficient of determination; RMSE, root mean squared error; XGBR, eXtreme gradient boosting regressor.

(BMs), lymph node metastases (LNMs), or soft tissue (ST), with the prostate bed considered ST. A second physician validated the segmentations.

Post-therapy imaging, calibration, and segmentation enabled Lu-PSMA dosimetry and the calculation of ADs in 130 segmented lesions, as detailed previously.¹⁸ CT images were resampled to match SPECT resolution using Lanczos interpolation in 3D Slicer, and Hounsfield Units (HUs) were converted to materials and densities based on the tables provided by Schneider. Activity and density maps from SPECT/CT were input into the GATE simulation platform (v9.2, Geant4 v11.0) to generate SPECT-based DRMs (Gy/s). The GATE simulation recorded the AD in a 3D matrix with a 3% statistical uncertainty. ADs for each VOI were calculated by multiplying the AD rate by region-based time-integrated activity coefficients (TIACs) and computed in Gy/GBq. These ADs were used as labelled data for the first objective of this study, which was to develop an ML-based dose prediction model using radio-dosimetrics.

For the second objective of this study, PET DRMs were generated through direct full MC simulations using the GATE v9.0 toolkit to acquire DFs for an ML-based model designed to predict lesion ADs in Lu-PSMA TRT. In this regard, the radioisotope voxelized source distribution and the patient's anatomical characteristics were simulated by importing patient PET and CT data into GATE. GATE features a tool called DoseActor, which records the AD within a specified volume as a 3D matrix.¹⁸ The Schneider tables were uti-

lized to convert CT HU into materials and densities.¹⁹ The simulations employed the MersenneTwister random number generator and included standard electromagnetic (emstandard opt3) and RadioactiveDecay physics processes. To ensure a low level of statistical uncertainty, 10⁹ primary events were simulated in each run, achieving an average statistical uncertainty of $\approx 3\%$ for every patient.

2.3 | Machine learning-based dose prediction using radiodosimetrics

2.3.1 | Feature extraction

After manual segmentation, image pre-processing, and feature extraction were carried out utilizing the LIFEx software (version 7.4.0). The extraction process followed the most up-to-date benchmark values outlined by the Image Biomarker Standardization Initiative (IBSI) to ensure consistency.^{20,21} Pre-processing involved spatial resampling to a uniform voxel size of $3 \times 3 \times 3$ mm³, absolute intensity rescaling, and discretization, with standardized parameters applied across all patients. DRM images were discretized into 64 gray levels using absolute intensity rescaling, while CT images were binned into 10 discrete levels within a fixed Hounsfield Unit (HU) range of -1000 to 3000 . RFs from CT images and DFs from the DRMs were then extracted. These included first-order statistical metrics

(e.g., mean, standard deviation), shape-based descriptors, and higher-order texture features derived from gray level co-occurrence matrix (GLCM), gray level run-length matrix (GLRLM), gray level size zone matrix (GLSZM), and neighbouring gray tone difference matrix (NGTDM).

A total of 243 features (RFs + DFs) were obtained from each volume of interest (VOI), comprising 119 RFs from CT and 124 DFs from DRM per pretreatment Ga-PSMA image dataset. Features were extracted from various categories, including First Order, GLCM, GLDM, GLRLM, GLSZM, NGTDM, Shape, and Shape2D. Alongside RFs capturing anatomical characteristics, CBs were incorporated into the model. Radiomics processing and the ML pipeline were implemented using Python V.3.11.5. Before selecting features, all extracted data underwent z-score normalization, as represented in Equation 1, to ensure a standardized training dataset with a mean of zero and a variance of one:

$$z = \frac{X - \mu}{\sigma} = \frac{X - \frac{1}{N} \sum_{i=1}^N (X_i)}{\sqrt{\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}}} \quad (1)$$

In this equation, X represents the radiodosimetric feature, μ denotes the mean of all radiodosimetric features, σ corresponds to the standard deviation (SD) of this mean, and N signifies the total number of radiodosimetric features.

2.3.2 | Feature selection

Overfitting is a major challenge in ML studies, especially when the number of extracted features exceeds the available sample size. To mitigate this issue and effectively reduce the dimensionality of the training dataset, multiple feature selection techniques were applied, including recursive feature elimination (RFE), mutual information (MI), Boruta, Elastic Net, and Least Absolute Shrinkage and Selection Operator (LASSO). For each method, selected features were further evaluated using Spearman's rank correlation with AD, and statistical significance was assessed by the Benjamini-Hochberg (B-H) correction, with a q -value < 0.05 considered statistically significant. This multi-method approach was employed to identify which techniques yield the most meaningful features and contribute to improved performance in ML-based dose prediction.

2.3.3 | Machine learning model construction

Five nonlinear ML regression algorithms were utilized to identify the optimal model: Extra Tree Regres-

sor (ETR), Random Forest Regressor (RFR), Gradient Boosting Regressor (GBR), eXtreme Gradient Regressor (XGBR), and Adaptive Boosting (AdaBoost) Regressor.

A series of experiments was conducted to assess the predictive capability of different feature categories, CT RFs, DRM DFs, and CBs, in estimating mean ADs. These experiments were integrated into ML models.

- Experiment 1 analysed selected CT RFs combined with selected DRM DFs.
- Experiment 2 incorporated selected CBs alongside selected CT RFs and DRM DFs.
- Experiment 3 expanded the approach by including all CBs in addition to selected CT RFs and DRM DFs.

Feature selection within each experiment was used to identify the most relevant predictors and determine the optimal ML algorithms.

2.3.4 | Machine learning model evaluation

To train and evaluate the model, leave-one-out cross-validation (LOOCV) was implemented to address the limitations of the small sample size. This method allowed the model to be trained on nearly all available data in each iteration, maximizing learning compared to other validation techniques.

To evaluate the predictive performance of the ML regression models, five metrics were computed: root mean squared error (RMSE), mean squared error (MSE), mean absolute error (MAE), coefficient of determination (R^2), and Spearman's rank correlation (ρ , q -value < 0.05). Since MAE, MSE, and RMSE are scale-dependent, they are suitable for comparing regression models within the same dataset. In contrast, R^2 quantifies the proportion of variance in the target variable explained by the input features. The MAE, MSE, and RMSE were expressed in Gy/GBq.

2.4 | Deep learning-based dose prediction

2.4.1 | Deep learning model construction

Prior to training, the dataset was preprocessed and standardized to match the model's input requirements. B-Spline interpolation was applied to downsample CT slices, ensuring they matched the spatial coordinates of the corresponding PET images. As a result, both CT and PET images were sampled to identical dimensions (168×168) and a uniform voxel size of $4.07 \text{ mm} \times 4.07 \text{ mm} \times 3 \text{ mm}$, while preserving anatomical alignment. Finally, CT and PET values were normalized to a range between

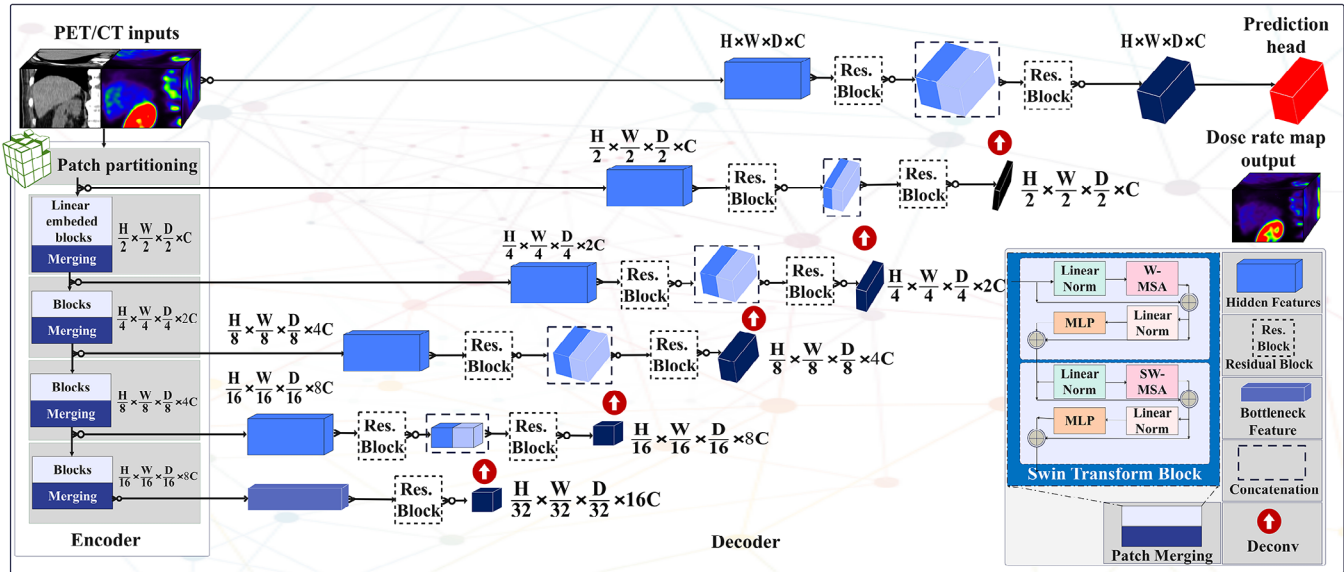


FIGURE 1 Overview of the proposed Swin UNETR-based architecture for voxel-wise dose rate map prediction from positron emission tomography/computed tomography (PET/CT) images. The input dual-channel PET/CT volume is partitioned into non-overlapping 3D patches and passed through a hierarchical Swin Transformer encoder consisting of four stages with progressive patch merging and feature embedding. The decoder reconstructs the spatial resolution through deconvolution and skip connections with the encoder. Residual blocks refine features at multiple levels, and a convolutional neural network (CNN)-based prediction head outputs the final 3D dose rate map. The Swin Transformer blocks utilize window-based multi-head self-attention (W-MSA) and shifted-window-MSA (SW-MSA) self-attention mechanisms for efficient feature representation.

0 and 1 using Equation 2:

$$X_{norm} = \frac{(X - X_{min})}{(X_{max} - X_{min})} \quad (2)$$

Moreover, as another preprocessing step, MONAI's data augmentation techniques, including random flipping, rotation, and intensity scaling, were applied to enhance model generalization. Specifically, transformations like RandFlipd, RandRotate90d, RandScaleIntensityd, and RandShiftIntensityd were utilized with predefined probabilities to introduce variations in the input data. These augmentations are commonly used for PET/CT datasets and were retained from our prior work.

For model development, the Swin UNETR architecture was employed, which leverages a Swin Transformer as its encoder, as outlined in the reference.²² This hierarchical vision transformer utilizes shifted windows to effectively capture both local and global features. Furthermore, it integrates a CNN-based decoder with skip connections across multiple resolutions, facilitating efficient feature propagation and reconstruction.

As shown in Figure 1, the architecture begins with dual-input PET/CT volumes, which are first passed through a patch partitioning module that divides the input volume into non-overlapping 3D tokens (patches) of size $2 \times 2 \times 2$. These patches are linearly projected and progressively downsampled through four hierarchical stages in the encoder, each stage halv-

ing the spatial dimensions and increasing the number of channels ($C \rightarrow 2C \rightarrow 4C \rightarrow 8C \rightarrow 16C$). The core of each encoder stage consists of Swin Transformer blocks that perform window-based multi-head self-attention (W-MSA) and shifted-window-MSA (SW-MSA) modules, as depicted in the figure's legend. The bottleneck representation is followed by a decoder path where deconvolution (upsampling) layers are performed at each stage to gradually reconstruct the spatial resolution. Skip connections between encoder and decoder blocks at matching resolutions enable effective fusion of low-level and high-level features. Residual blocks are incorporated throughout the architecture to enhance gradient flow and representation power. Finally, the concatenated features are passed through a CNN-based prediction head to generate the 3D DRM output. This design enables end-to-end voxel-wise DRM prediction from multimodal PET/CT images.

In previous work, an SSL framework for PET/CT segmentation was developed, leveraging a Swin UNETR model trained on 752 images, 652 for SSL and 100 labelled images for fine-tuning.¹⁷ This SSL approach employed three proxy tasks: image rotation prediction, image volume inpainting, and contrastive coding, allowing the model to learn meaningful anatomical representations. During pretraining, only the encoder was trained; it was subsequently integrated with the decoder for fine-tuning on labelled data. This SSL approach enabled the model to benefit from a large amount

of unlabelled data, significantly improving performance during the fine-tuning stage.

In the current study, the Swin UNETR architecture remains structurally unchanged to enable the use of pretrained SSL weights from the previous segmentation task, and it was further fine-tuned on 30 PET/CT images to predict dose distributions.¹⁷ This transfer learning approach leveraged the learned anatomical features from PET/CT images, improving the model's performance on the new task. However, the network output and training objective were modified to suit the new task of voxel-wise DRM prediction. Specifically:

- The output was adapted from a 12-channel segmentation map to a single-channel continuous DRM.
- The loss function was changed from Dice loss (used for segmentation) to MSE, which is appropriate for regression tasks.
- No additional architectural or augmentation changes were introduced, as the pretrained model and input modality (PET/CT) were consistent between tasks, enabling efficient fine-tuning through transfer learning.

This transfer learning approach leveraged the anatomical knowledge already encoded in the model, allowing the network to generalize more effectively to the new DRM prediction task.

The study was carried out in six distinct phases. The first three phases involved the use of nnU-Net with PET-only input, nnU-Net with PET/CT inputs, and ResNet152 with PET/CT inputs—all well-established networks commonly employed in medical image processing. These phases provided a baseline for comparing the performance of the transformer network with traditional convolutional neural networks (CNNs). In the fourth and fifth phases, the Swin UNETR model was trained from scratch using PET-only and PET/CT inputs, respectively, without the use of self-supervised pre-training. The final phase involved fine-tuning the pretrained Swin UNETR model using the SSL weights from the Ga-PSMA segmentation task, now applied to the DRM prediction dataset based on dual-modality PET/CT inputs.¹⁷

The network was trained to predict MC-simulated DRMs. The dataset was divided into five equal subsets for 5-fold cross-validation, ensuring uniform data distribution across all folds. During fine-tuning, the AdamW optimizer (learning rate = $1e-4$, weight decay = $1e-5$) was employed, along with a warm-up cosine annealing learning rate schedule, to handle sparse gradients and enhance convergence. A dropout rate of 0.2 was applied uniformly across the model to reduce overfitting and improve generalization. In addition, early stopping was employed based on validation loss, with a patience of 50 epochs to prevent overtraining and ensure model convergence. The model was trained for a total of 1000

epochs using the MSE loss function to minimize the squared differences between predicted and true values. By minimizing MSE, the model is encouraged to make predictions that are as close as possible to the actual dose distribution. Model implementation was carried out using PyTorch 1.13 and MONAI 0.9. All experiments were conducted on an NVIDIA RTX 3090 GPU with 24 GB of memory.

2.4.2 | Deep learning model evaluation

The dose maps generated by DL and those calculated via MC simulations were subjected to a quantitative comparison using multiple evaluation metrics. These metrics include voxel-wise RMSE, normalized root mean square error (NRMSE), MAE, mean error (ME), relative error (RE), relative absolute error (RAE), and peak signal-to-noise ratio (PSNR). Additionally, a Gamma index evaluation was conducted using the `gamma_index` function from the GateTools Python package, with the criteria of a 4.072 mm distance to agreement (DTA) and a 1% dose difference (DD). The Gamma pass rates and corresponding Gamma index map were computed to assess the agreement in dose distribution between the DL-generated and MC reference DRMs. The pass rates were calculated using Equation 3 as follows:

$$\text{Pass rate (\%)} = \frac{\text{Voxels number with } \gamma \leq 1}{\text{Total voxels number}} \times 100 \quad (3)$$

3 | RESULTS

3.1 | Machine learning-based dose prediction using radio-dosimics

The predictive performance of five non-linear ML-based regression algorithms was assessed for estimating ADs within lesions using pretreatment Ga-PSMA DRMs and Ga-PSMA CT scans, following the first treatment cycle of Lu-PSMA TRT. Reference dose values were obtained from MC simulations, and the ground truth was considered for all comparisons with the predicted dose values generated by the models.¹⁸

To evaluate the impact of feature selection on model performance, several methods were tested, including RFE, MI, Boruta, Lasso, and Elastic Net. Among these, RFE consistently outperformed the others, yielding the most informative and predictive feature subsets across all experiments. Accordingly, the selected features for each of the three experiments (E1–E3) based on RFE are outlined in Table 1. For completeness, the features selected by other methods and their corresponding model performances are reported in Tables S1–S8 in the Supplementary Tables.

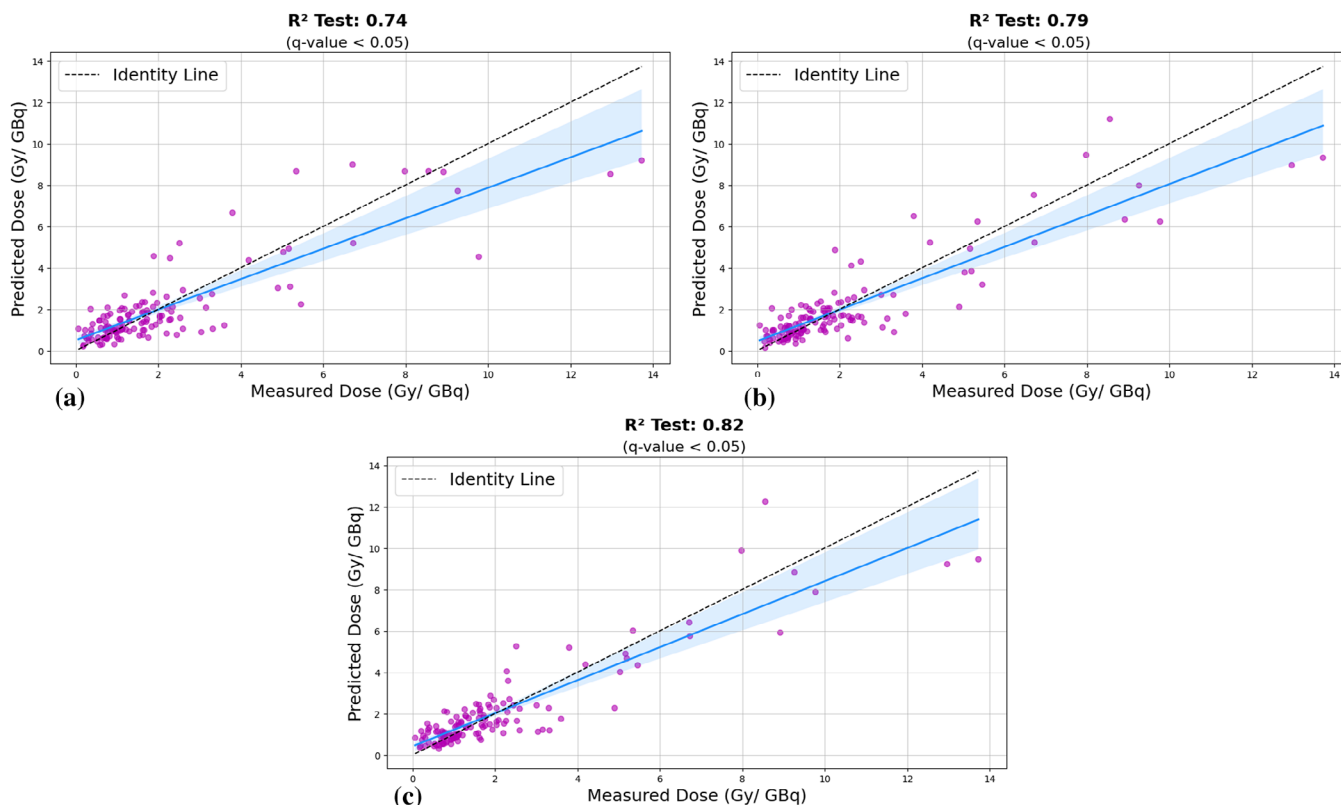


FIGURE 2 R^2 values of the best predictive models across three experiments: XGBR in Experiment 1 (a), XGBR in Experiment 2 (b), and extra tree regressor (ETR) in Experiment 3 (c), illustrating predicted versus measured doses (Gy/GBq).

As summarized in Table 1, E1 selected 7 DFs and 3 RFs from DRMs and CT images, respectively. E2 included PSA and ALK from CBs in addition to the E1 features. E3 incorporated all statistically significant CBs ($q < 0.05$), including PSA, ALP, CA, P, HCT, and CR, together with the E1 features.

Table 2 summarizes the performance of the best predictive algorithms in each experiment. The best model in E1 was XGBR, which also achieved the best performance in E2. In E3, the best model was ETR. The R^2 values across the experiments, shown in Table 2 and illustrated in Figure 2 (panels a–c), demonstrate a progressive improvement in predictive accuracy. In E1, XGBR achieved a moderate R^2 , reflecting a reasonable fit between predicted and measured doses. The R^2 score improved in E2 with the addition of CBs, highlighting the benefit of including these biomarkers in the model. The highest R^2 value was observed in E3, where the ETR model outperformed the others, indicating the most accurate dose predictions.

Other performance metrics, including MAE, MSE, and RMSE, followed a similar trend, with improvements as whole CBs ($q\text{-value} < 0.05$) were added, leading to more stable and reliable outcomes. The Spearman's rank correlation values further supported the models'

effectiveness, showing strong associations between the predicted and measured doses in each experiment.

Figure 3 presents a Spearman's rank correlation heatmap depicting the relationships between the selected features in E3, their self-correlations, and their correlation with the target variable (dose). Correlation values range from -1 to +1, with values closer to +1 indicating a strong positive correlation, and values closer to -1 indicating a strong negative correlation.

ML models are often referred to as “black-box models” due to their complexity and lack of transparency in their decision-making processes. To improve interpretability and build trust, this study utilized post hoc explainable artificial intelligence (XAI) techniques, specifically, Shapley Additive Explanations (SHAP) and local interpretable model-agnostic explanations (LIME) interpretability tools. SHAP uses the Shapley value from game theory to assign each feature's contribution to the model's output, helping to clarify individual predictions. In contrast, LIME is a model-independent method that explains complex models by creating simplified surrogate models based on the local neighbourhood of the analysed instance.²³ Unlike ante hoc XAI methods, which design interpretability into the model architecture itself, SHAP and LIME are post hoc tools that provide insights after the model has been trained.^{24,25}

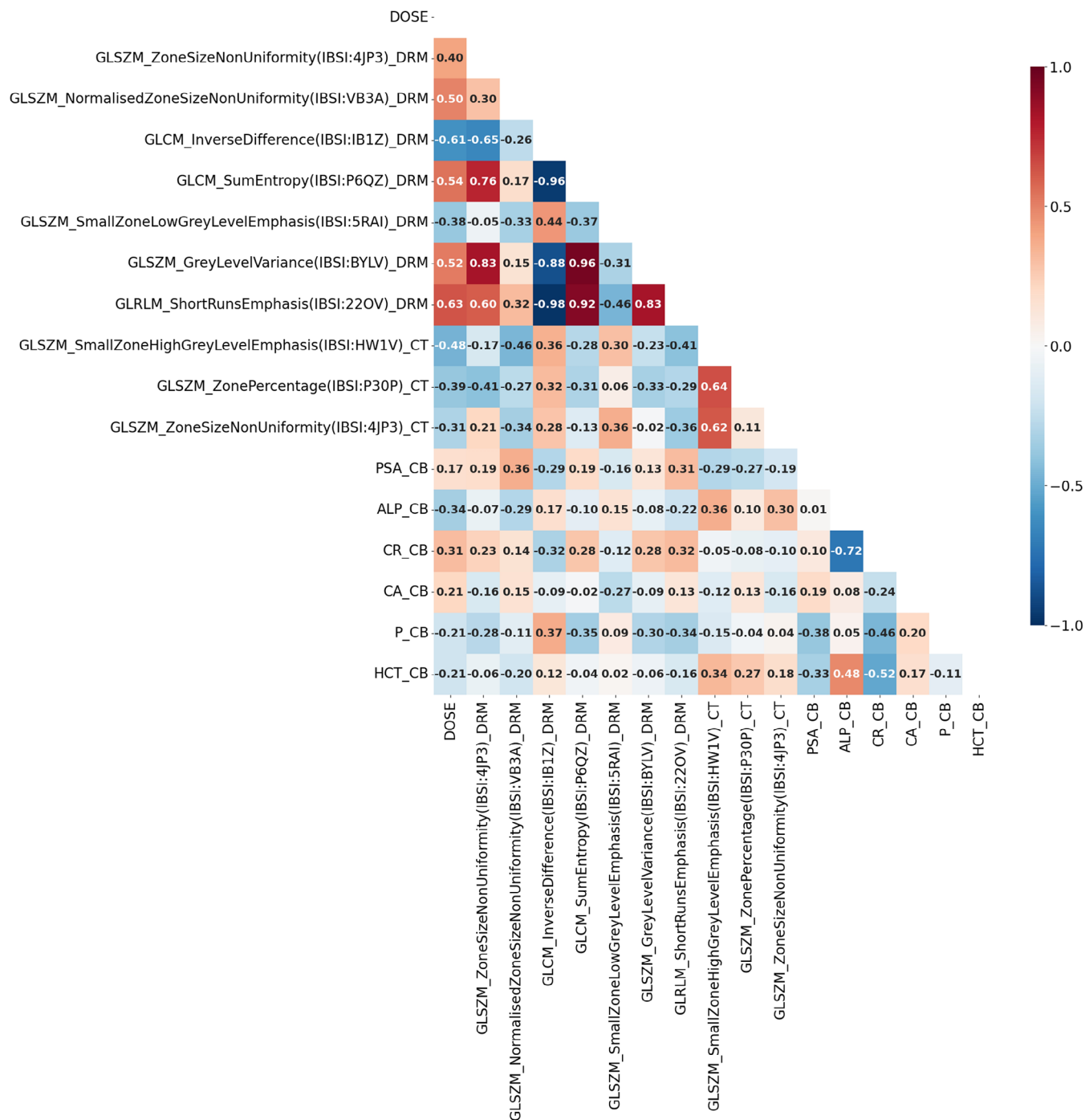


FIGURE 3 The Spearman's rank correlation heatmap of the selected features in the last experiment (E3).

The SHAP model visualization in Figure 4a illustrates the influence of key features on the prediction outcomes across experiment E3. In these plots, pink points represent features with a positive impact (increasing the predicted value), while blue points indicate a negative effect (decreasing the predicted value). Prominent features, such as “*GLCM_SumEntropy_DRM*” and “*GLSZM_NormalisedZoneSizeNonUniformity_DRM*” in Figure 4a, show both positive and negative

effects, with their influence varying across different observations. Intermediate features like “*GLSZM_SmallZoneLowGreyLevelEmphasis_DRM*” display a more limited effect, with points clustered around the central value. Less influential features, such as “*GLCM_InverseDifference_DRM*” and “*HCT_CB*,” exhibit minimal impact, as indicated by the points clustering near zero. According to Figure 4a, PSA and ALP are ranked as the 5th and

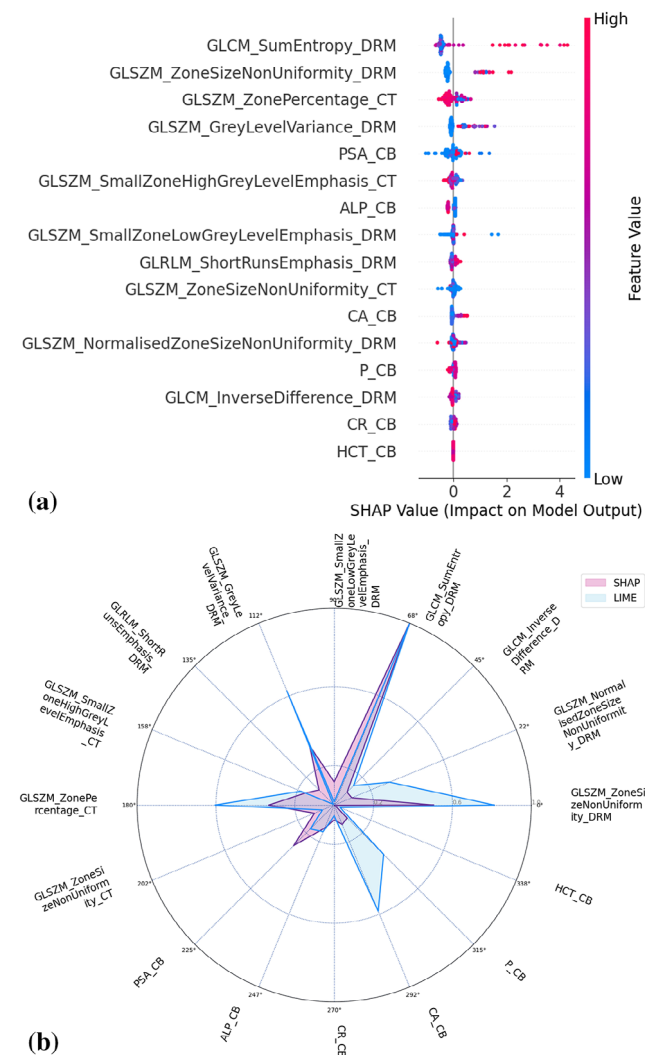


FIGURE 4 Feature importance for experiment E3. (a) Shapley Additive Explanations (SHAP) summary plot showing feature impact on the best model, extra tree regressor (ETR). (b) Radar plot comparing SHAP (pink) and LIME (blue) importance, highlighting differences in the methodologies' assessments of feature contributions.

7th most important features in the best model of E3 (ETR).

Figure 4b shows radar plots for experiment E3, illustrating feature importance in the best models based on SHAP and LIME. The radar plot visually displays feature importance in a circular, multi-dimensional format. Features with greater influence extend farther from the centre, with points closer to the outer edge indicating higher importance. The plot's outline shows how importance values compare across features. Each chart maps normalized importance values on a circular grid, with axes representing individual features. SHAP and LIME values are normalized to a range of 0 to 1, allowing for direct comparison of feature importance across both methods. The coloured areas indicate feature contributions, while the lines outline the overall importance distribution. For

instance, in Figure 4b, *GLCM_SumEntropy_DRM* is the most influential feature according to both SHAP and LIME, though SHAP assigns it a slightly higher importance. However, features like *PSA* and *GLSZM_SmallZoneLowGreyLevelEmphasis_DRM* exhibit noticeable disparities, with SHAP attributing more influence to them compared to LIME. The differences highlight SHAP's global feature importance and LIME's local perturbation approach, offering complementary insights into feature relevance.

3.2 | Deep learning-based dose prediction

Table 3 summarizes the performance of six models in DRM prediction using either PET-only or combined PET/CT inputs. The baseline models include nnU-Net (with PET and PET/CT) and ResNet101 (with PET/CT), while the advanced transformer-based models include Swin UNETR (with PET and PET/CT) and Swin UNETR with SSL pre-training (with PET/CT).

Across both nnU-Net and Swin UNETR experiments, incorporating PET/CT inputs consistently improved model performance over PET-only inputs. Notably, Swin UNETR with PET/CT inputs outperformed its PET-only counterpart across nearly all metrics, demonstrating the value of multimodal input. Similarly, nnU-Net showed clear improvement when PET/CT was used instead of PET alone.

The best performance was achieved by Swin UNETR with SSL pre-training and PET/CT inputs, showing superior results in terms of R^2 score, error metrics, PSNR, and gamma pass rate. These best values are underlined in Table 3.

Figure 5 depicts a patient with mCRPC, presenting the CT scan as an anatomical reference alongside MC DRM, DL DRM, and a 3D Gamma evaluation, using MC dose maps as a reference standard. The Gamma index map highlights the region with a few rejected voxels (Gamma value > 1).

4 | DISCUSSION

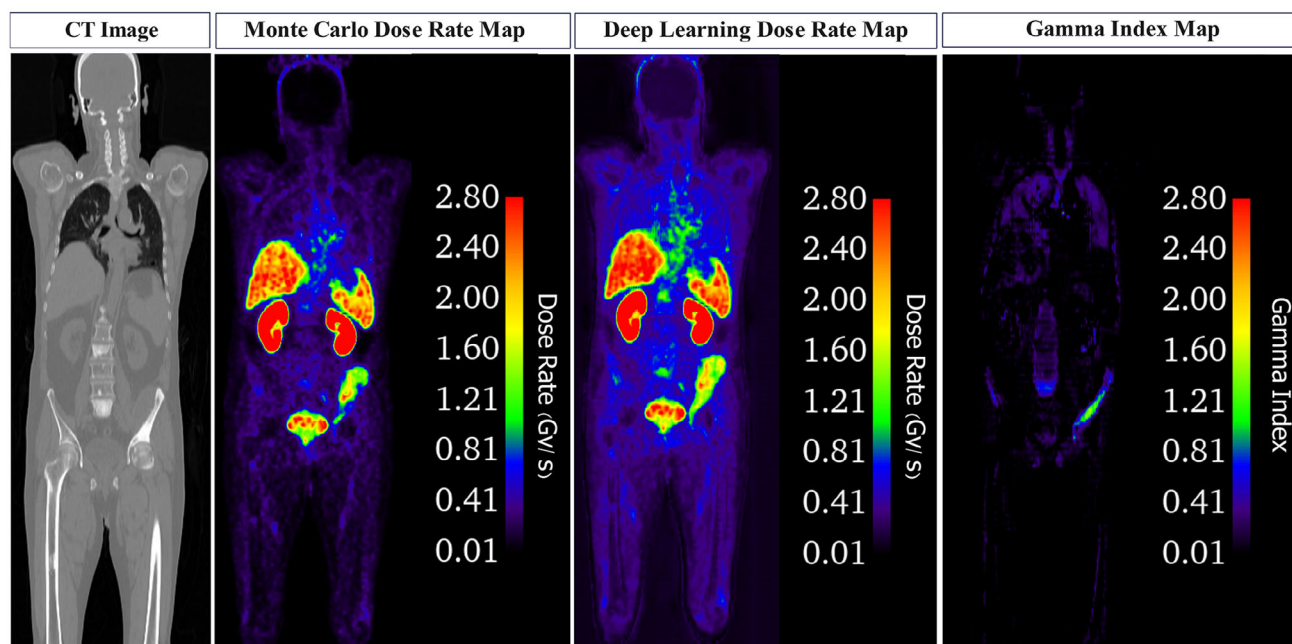
Early and accurate prediction of tumour and organ ADs in patients with mCRPC is critical for optimizing therapeutic outcomes.²⁶ This information enables personalized treatment planning by allowing pretherapy adjustment of TRT doses to maximize tumour irradiation while minimizing toxicity to healthy tissues.²⁷ While Ga-PSMA PET-derived quantitative metrics, such as SUV, have demonstrated potential in reflecting PSMA expression in metastatic tumors,²⁸ their correlation with Lu-induced tumour doses and clinical response remains limited.^{28–31}

TABLE 3 Performance comparison of six models in dose rate map (DRM) prediction using either PET-only or positron emission tomography/computed tomography (PET/CT) inputs.

Model	R ² Score	ME	RE (%)	MAE (Gy/s)	RAE (%)	RMSE (Gy ² /s)	NRMSE	PSNR (Gy/s)	Gamma pass rate
nnU-Net PET	0.86	0.144	0.332	0.048	0.476	0.143	0.007	46.36	89.32%
nnU-Net PET/CT	0.93	0.097	0.232	0.042	0.445	0.087	0.004	47.79	94.22%
ResNet101 PET/CT	0.64	0.014	0.583	0.085	1.242	0.156	0.011	34.90	87.05%
Swin UNETR PET	0.89	0.055	0.264	0.057	0.561	0.111	0.005	47.14	93.14%
Swin UNETR PET/CT	0.94	<u>0.013</u>	0.199	0.037	0.343	0.087	0.004	49.66	98.97%
SSL pre-trained + Swin UNETR PET/CT	<u>0.97</u>	0.014	<u>0.140</u>	<u>0.026</u>	<u>0.301</u>	<u>0.052</u>	<u>0.003</u>	<u>52.21</u>	99.08%

*Abbreviations: MAE, mean absolute error; ME, mean error; NRMSE, normalized root mean square error; PSNR, peak signal-to-noise ratio; R² Score, coefficient of determination; RAE, relative absolute error; RE, relative error; RMSE, root mean square error.

The baseline models include nnU-Net and ResNet101, while Swin UNETR and SSL pre-trained Swin UNETR represent transformer-based approaches.

**FIGURE 5** Computed tomography (CT) image, Monte Carlo (MC)-derived dose rate map (DRM), deep learning (DL)-derived DRM, and Gamma index map in the coronal views for a case with a 98.65% pass rate using the self-supervised learning (SSL) pretrained + Swin UNETR model. Voxels with Gamma values ≤ 1 met the pass criteria.

In oncology, AI-driven models have been utilized to support tumour detection from imaging data, stratify patient risk using histopathology or RFs, predict survival outcomes and recurrence risk, and estimate or predict absorbed radiation doses.^{14,32,33} The earliest application of AI for dose estimation was reported in 2011 in the context of EBRT.³⁴ In the following years, a growing number of DL-based techniques were introduced for dose prediction, reflecting the field's rapid advancement.^{6,35} Xue et al.⁷ advanced their approach to dose estimation in patients with mCRPC by transitioning from a random forest algorithm⁸ to a DL-based method, achieving improved performance and predictive

capability. Plachouris et al.¹⁵ developed an ML-based predictive dosimetry model for [¹⁷⁷Lu]Lu-DOTATATE therapy. By combining RFs and DFs extracted from pre- and post-treatment imaging, the model achieved accurate AD predictions for organs, supporting the potential of radiodosimetrics in personalizing TRT and guiding dose escalation decisions.

In our previous work involving standard-dose Lu-PSMA therapy, we developed a predictive model to estimate ADs in kidneys and tumour lesions using pretherapy Ga-PSMA PET/CT-derived RFs and CBs.¹⁸ This model achieved a renal dose prediction with an RMSE of 0.11 Gy/GBq and an R² of 0.87, while lesion

dose prediction reached an RMSE of 1.04 Gy/GBq and an R^2 of 0.77 using the optimal configuration. While PET imaging provides valuable insight into radio-tracer distribution within tissues, MC-derived AD maps provide a more accurate representation of the actual radiation dose deposited. These physical processes introduce spatial and textural variations in the DRMs that are not linearly or uniformly represented in the activity distributions from PET. This distinction is crucial for radionuclides such as Lu-177, whose short-range beta emissions lead to highly localized energy deposition, with limited gamma radiation contribution.¹⁵

Building on this foundation, the primary objective of the present study was to assess the feasibility of integrating DFs extracted from pretreatment Ga-PSMA DRMs, in combination with Ga-PSMA CT-based RFs, and routinely available baseline CBs, to identify radiodosiomic signatures capable of predicting lesion-specific ADs in Lu-PSMA TRT for 20 patients with mCRPC. To ensure reliable model performance, this study utilized Ga-PSMA DRMs generated through MC simulations as reference data. These DRMs served as ground truth for identifying areas receiving optimal radiation doses, allowing for potential refinement of treatment planning to enhance therapeutic effectiveness while reducing adverse effects. To support this goal, we evaluated various ML algorithms for their ability to predict AD, aiming to personalize TRT strategies for patients with mCRPC.

Although post-therapy SPECT of Lu-PSMA distribution is the standard modality for clinical dosimetry, this study focuses on pretreatment image-based prediction to inform personalized treatment planning before RLT. In this context, the current study used Ga-PSMA PET/CT imaging for DRM generation, supported by several key considerations: (1) PET offers higher spatial resolution and quantification accuracy than SPECT, especially for small lesion visualization and segmentation; (2) Ga-PSMA PET/CT is routinely acquired as part of diagnostic staging in patients with mCRPC and is the only modality available prior to therapy, making it ideal for prospective modelling; (3) prior studies have shown that PET uptake correlates with post-therapy AD, supporting its predictive value^{8,28} (4) PET-based MC simulations provide practical training data for ML, especially when SPECT dosimetry is unavailable or limited; and (5) PET-based DRM modelling aligns with the clinical workflow and enables real-time, patient-specific dosimetry predictions. Therefore, although PET is not conventionally used for dose calculation, it serves as a powerful tool for learning and predicting spatial dose patterns before therapy. Moreover, while PET images and DRMs may appear superficially similar due to the short range of Lu-177 beta emissions, the MC simulation process introduces biologically and physically meaningful spatial heterogeneity that supports the rationale for extracting distinct DFs.

Furthermore, the increasing adoption of RPT highlights the urgent need for simplifying dosimetry workflows and reducing the reliance on multiple post-therapy image acquisitions. Just as pretreatment planning revolutionized EBRT, leveraging routinely acquired pretherapy PET/CT for dose estimation could streamline clinical decision-making in RPT. These early radiodosiomic studies, while limited in scale, are essential for raising awareness within the community about the untapped potential of PET/CT imaging, from conventional SUV metrics to advanced high-order features, for informing dose-guided therapy.

Our results suggest that the combination of PET-derived DFs, CT-derived RFs, and all CBs with $q < 0.05$, as identified in E3 (Table 2), represents the optimal feature set for the ETR model. This combination enables more accurate prediction of post-therapy lesion ADs, achieving an RMSE of 0.67 Gy/GBq and an R^2 of 0.82. Incorporating CBs markedly improves the predictive accuracy, emphasizing the complementary value of systemic disease indicators for modelling lesion-level radiation dose. The increase in Spearman's rank correlation from 0.70 in E1 to 0.79 in E3 further supports the strengthened association between predicted and measured ADs. These findings support the hypothesis that combining PET-derived DFs, CT-based RFs, and relevant CBs yields a synergistic effect, allowing for more precise lesion-level dose estimations.

While the marginal performance improvement when combining radiomics and dosiomics must be interpreted with caution due to the sample size, the consistent selection of certain features across folds suggests that they contain non-redundant spatial information, particularly in voxel-wise analyses of heterogeneous lesions. This finding supports the hypothesis that MC-based dose patterns encode clinically relevant features that are not fully captured by PET activity distributions alone.

MC-based DRMs provide the most accurate reference for training ML models, being particularly suitable for scenarios involving heterogeneous activity distributions and media. However, their computational demands and complex setup processes limit their practical use in routine clinical treatment planning.^{2,36,37} While fully supervised DL approaches offer high accuracy and efficiency for dose estimation, they risk overfitting when the training dataset is insufficient relative to the model's complexity. To address this challenge, Zhang et al.^{9,38} proposed a semi-supervised learning framework to personalize dosimetry in TRT using pre-therapy Ga-PSMA PET/CT images, mitigating the issue of limited post-therapy SPECT data. By adopting the FixMatch method for regression, their approach demonstrated improved organ dose prediction accuracy in an in-silico study using synthetic data and MC simulation. Mansouri et al.³⁹ recently leveraged a UNETR DL architecture to predict radiation dose for [¹⁷⁷Lu]Lu-DOTA-TATE TRT,

demonstrating superior accuracy and generalizability compared to conventional approaches.

The second objective of the current study was to develop a DL model to facilitate the generation of DRM data. In this regard, a Swin UNETR architecture, a transformer-based model, was employed. This model was initially pre-trained on 652 unlabelled Ga-PSMA PET/CT images from a prior segmentation study utilizing SSL principles, enabling it to learn rich feature representations. It was then fine-tuned using pseudo-labelled DRMs derived from 30 MC-based simulations, including 10 additional patients added specifically for this task.

Unlike the ML model, which relied on accurate post-therapy dosimetry data from Lu-PSM SPECT/CT scans available for only 20 patients, the DL model predicted voxel-wise DRMs directly from pretreatment Ga-PSMA PET/CT images. The pseudo-labels for DL training were generated solely from MC simulations based on pretherapy imaging, eliminating the need for post-therapy dosimetry data. This key distinction allowed the addition of 10 patients without post-therapy scans due to clinical limitations, thereby expanding the DL training dataset. This expansion improved the spatial generalizability of the DL model without affecting the ML analysis.

By combining extensive pre-training with task-specific fine-tuning, this approach enables accurate voxel-wise DRM prediction and offers a promising tool for personalized TRT planning in patients with mCRPC.

The findings in Table 3 underscore the strength of transformer-based architectures, particularly Swin UNETR, for voxel-wise DRM prediction, with notable improvements observed when PET/CT inputs are used. Importantly, the integration of SSL pre-training produced the best performance across all metrics, even outperforming standard Swin UNETR. This highlights not only the architectural advantage of the Swin Transformer in capturing long-range dependencies but also the value of pre-training the encoder on a larger dataset. Although the current DRM prediction task was limited to only 30 labelled cases, the successful transfer of encoder weights pre-trained on 652 unlabelled cases demonstrates robust cross-task generalizability.¹⁷

This is particularly impactful given the significantly higher resource demands of DRM annotation compared to segmentation, which requires time-consuming MC simulations. Therefore, the ability to leverage prior segmentation-based SSL pre-training mitigates data scarcity and enhances prediction accuracy, reinforcing the practical value of SSL in dosimetry pipelines. Moreover, the superior performance of Swin UNETR over the ResNet101-based and nnU-Net models, commonly used benchmarks in medical imaging, suggests that hierarchical vision transformers are better suited for capturing the complex spatial patterns inherent in dose distribution tasks.

The SSL + Swin UNETR model achieved the highest R^2 (0.9717), lowest RE (0.14%), and lowest NRMSE (0.00303), along with the highest gamma pass

rate (99.08%), indicating strong spatial and dosimetric agreement with MC. Figure 5 qualitatively illustrates these findings. The DL DRM closely matches the MC DRM, accurately capturing high-dose regions. The Gamma index map shows minimal voxel rejection ($\gamma > 1$), confirming high agreement between MC and DL predictions. These results highlight the potential of SSL + Swin UNETR as a rapid and accurate alternative to MC for patient-specific DRM prediction.

Given the limited dataset and the architectural complexity of the Swin UNETR model, addressing the risk of overfitting was a critical aspect of this study. Several complementary regularization strategies were implemented to enhance model generalizability and robustness. Data augmentation techniques, including random rotations, intensity shifts, and flipping, were applied to introduce variability in anatomical presentation and image intensity. Early stopping was employed to terminate training once validation loss plateaued, preventing overfitting, while dropout layers were incorporated to reduce neural co-adaptation. Cross-validation was also used to monitor generalization performance. Furthermore, the model benefited from self-supervised pretraining using 652 unlabelled Ga-PSMA PET/CT scans, which facilitated transfer learning and more stable feature extraction during fine-tuning.¹⁷ Although Bayesian optimization was not explored in the current work, it represents a promising strategy for systematic hyperparameter tuning. Collectively, these measures were crucial for achieving robust voxel-wise DRM prediction from limited labelled data in patients with mCRPC.

In summary, this study demonstrates the potential of radiodosimics and DL-based models for personalized dosimetry in patients with mCRPC. By integrating PET-derived DFs, CT-based RFs, and CBs, we achieved accurate lesion-level dose predictions using MC-derived ground truth. The successful transfer of SSL pre-training from segmentation to dose prediction highlights its utility in overcoming the challenge of limited DRM data. Nonetheless, several limitations warrant consideration.

Despite the encouraging results, several limitations must be acknowledged. For the ML component of this study, model development was based on 130 lesion-level data points derived from 20 patients who underwent full post-therapy Lu-PSMA dosimetry, enabling AD prediction using radiodosimic features. However, the modest sample size may limit the generalizability of these findings, and further external validation is necessary to confirm the added predictive value and non-redundancy of DFs relative to radiomic ones.

By contrast, the DL model was designed to predict voxel-wise DRMs directly from pretreatment Ga-PSMA PET/CT images, thereby removing the need for post-treatment dosimetry data. This distinction enabled us to include 10 additional patients, raising the total to 30, for DL model training. However, expanding beyond this number was constrained by the computational burden

of MC simulations needed to generate DRM ground truth data. To mitigate the limited sample size, we leveraged SSL using pretrained encoder weights from 652 unlabelled PET/CT scans, enabling effective fine-tuning on the 30-patient dataset. Nonetheless, both the ML and DL models were developed using single-centre data, limiting their generalizability. External validation using independent, multi-centre datasets is essential to assess model robustness across varied clinical settings and support potential clinical translation. Future studies should focus on collaborative efforts to validate the models across different institutions, imaging protocols, and scanner types, and explore harmonization techniques to ensure clinical applicability.

Moreover, the current analysis focused solely on baseline PET/CT imaging and biomarker data obtained prior to therapy, and therefore did not incorporate AD or imaging information from subsequent treatment cycles. This restricts the evaluation of longitudinal changes in dose-response relationships and limits insight into cumulative or delayed effects, which are important for comprehensive treatment planning and response assessment.

Another limitation of this work involves the uncertainties associated with quantitative PET/CT imaging and feature extraction. These include errors introduced by scatter and attenuation correction, segmentation accuracy, PVE, and voxel discretization. Although standardized acquisition protocols and scanner calibration were used, PET/CT data are subject to resolution limitations and quantification variability that can propagate into the extracted radiomic and DFs. Manual lesion segmentation was conducted by a certified nuclear medicine physician, with a second expert reviewing the results, and lesions < 4 mL were excluded to reduce PVE.

Nevertheless, the delineation of lesions with diffuse or ill-defined boundaries remains inherently subjective and susceptible to inter-observer variability, potentially affecting feature stability and model performance. Additionally, to limit voxel-wise inconsistencies, image preprocessing steps such as voxel size resampling, intensity normalization, and fixed bin discretization (64 bins for DRMs and 10 bins over HU for CT) were applied. While these measures helped standardize input data, the residual variability in PET/CT quantification remains a challenge. Future work should explore automated or semi-automated segmentation techniques (e.g., DL-based contouring), implement partial volume correction algorithms, and assess feature robustness through test-retest or phantom studies to quantify and minimize these sources of error.

Furthermore, different feature extraction parameters, or selection methods were not explored, which may have influenced the identified feature importance and outcomes. Finally, employing simplified posttherapy imaging protocols, such as hybrid SPECT-planar imaging or

limited planar time-point acquisition for kinetic modelling, may introduce additional uncertainties into dosimetric calculations. Despite these constraints, this study lays the groundwork for AI-driven dosimetry frameworks applicable to broader clinical contexts.

5 | CONCLUSION

Effective personalized dosimetry in Lu-PSMA RLT relies on accurate pretherapeutic prediction of ADs. This study presents a radio-dosimetrics framework combining SSL and a Swin UNETR model to achieve this goal. By leveraging pseudo-label generation, the proposed method enables accurate, patient-specific AD estimation from pretherapy PET/CT images, even with limited ground-truth dose data. Once trained on MC-derived DRMs, the model achieves near-gold-standard accuracy while substantially reducing computational time and complexity. The proposed approach holds strong potential for clinical translation, offering a rapid and scalable solution for individualized TRT planning. Future validation on large, multi-institutional datasets is recommended to support its routine implementation in clinical practice.

ACKNOWLEDGMENTS

This work was supported by the School of Medicine, Iran University of Medical Sciences (IUMS), Grant 25160. Additionally, this work has been supported by the Simorgh Supercomputer at Amirkabir University of Technology, under Contract No. ISI-DCE-DOD-Cloud-900808-1700. We acknowledge helpful discussions with Hanieh Sadat Jozi regarding plotting the Gamma map.

CONSENT TO PARTICIPATE/PUBLICATION

The Ethics Committee of the Iran University of Medical Sciences (IUMS) approved this study under ethics approval IR.IUMS.FMD.REC.1401.505, with the patient's written informed consent. Informed consent was waived by the ethics committee (IR.IUMS.FMD.REC.1401.505).

CONFLICT OF INTEREST STATEMENT

The author declares that there are no conflicts of interest.

DATA AVAILABILITY STATEMENT

The datasets are not publicly available due to ethical reasons but can be requested from the corresponding author (pgeramifar@tums.ac.ir) or the first author (yazdani.el@iums.ac.ir).

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How to cite this article: Yazdani E, Neizehbaz A, Karamzade-Ziarati N, et al. Transforming [¹⁷⁷Lu]Lu-PSMA-617 treatment planning: Machine learning-based radiodosimetrics and swin UNETR using pretherapy PSMA positron emission tomography/computed tomography (PET/CT). *Med Phys.* 2025;52:e70030. <https://doi.org/10.1002/mp.70030>